

Member Contributions — Group 52 (Super Mario Game)

Course: CS202 - C++ Programming / Object-Oriented Design

Class: 25A01

Group: 52

This document is the same data as [Member_Contributions.xlsx](#) (the TA-facing spreadsheet, in the format used by the group's other CS202 project) rendered as Markdown/PDF for readability — both are generated from one source so they cannot drift apart.

Summary

Number of students	2
Number of tasks	120
Number of task hours	223.5
Number of Git commits	316
Max student percentage	0.5
Project score	10

Do not edit the grey cells in the spreadsheet edition. TAs enter the score of all PAs in the project score and get the individual scores; students may enter an estimated project score to see how percentages affect it. [Percent](#) is set to 0.5/0.5 for both members — the raw Tasks/Hours/Git percentages below inform that judgment rather than dictate it, the same convention the reference spreadsheet uses.

No	Student ID	Full name	Tasks	Tasks %	Task Hours	Hours %	Git Commits	Git %	Percent	Score (Student)
1	25125083	Nguyễn Đình	75	62.5%	143.5	64.2%	258	81.6%	0.5	10

No	Student ID	Full name	Tasks	Tasks %	Task Hours	Hours %	Git Commits	Git %	Percent	Score (Student)
		Minh Huy								
2	25125084	Trần Gia Huy	45	37.5%	80.0	35.8%	58	18.4%	0.5	10

Note on the gap between the Git % column and the Tasks/Hours % columns: Member A's git identity carries a large share of small fix/doc/log commits (258 of 316) accumulated across the whole session history, while task count and estimated hours — a better proxy for actual design/implementation effort — split closer to 62/38. Both are shown rather than only the more flattering one.

Task-by-task breakdown

A task should be done by only 1-2 students. Each member worked far more than the minimum 5 tasks the reference template asks for.

No	Student ID	Full name	Task Description	Hours	Evidence
1	25125083	Nguyễn Đình Minh Huy	Configured CMake (FetchContent) for SFML 3.0.2 + Dear ImGui/ImGui-SFML and set up the project directory structure	2	see docs/ Group52_04/52.md
2	25125083	Nguyễn Đình Minh Huy	Drafted the v2.0 project spec (110 features), UML class diagrams and the implementation plan (TASKS.md, TASK_DIVISION.md)	3	see docs/ Group52_04/52.md
3	25125083	Nguyễn Đình Minh Huy	Wrote clean C++ base skeletons for managers, physics and entities	1.5	see docs/ Group52_04/52.md
4	25125083	Nguyễn Đình Minh Huy	Implemented MathUtils (vector ops, clamp, interpolation helpers)	1	see docs/ Group52_04/52.md

No	Student ID	Full name	Task Description	Hours	Evidence
5	25125083	Nguyễn Đình Minh Huy	Built GameStateManager/IGameState state-stack pattern and the Game singleton main loop at a fixed 60Hz timestep	3	see docs/ Group52_04/52.md
6	25125083	Nguyễn Đình Minh Huy	Implemented AABB collision primitives and a dynamic SpatialHash broadphase grid	2.5	see docs/ Group52_04/52.md
7	25125083	Nguyễn Đình Minh Huy	Built the PhysicsEngine update pipeline (gravity, gravity zones/water viscosity, axis-separated tile resolution)	2.5	see docs/ Group52_04/52.md
8	25125083	Nguyễn Đình Minh Huy	Implemented CollisionDetector/ CollisionResolver (solid tiles, conveyors, stomps/damage, player-vs-player bounces)	3	see docs/ Group52_04/52.md
9	25125083	Nguyễn Đình Minh Huy	Restructured Entity attributes to protected with friend-class access and read-only getters	1	see docs/ Group52_04/52.md
10	25125083	Nguyễn Đình Minh Huy	Implemented the IPlayerState decorator pattern (StarDecorator, MegaDecorator) with timers	2	see docs/ Group52_04/52.md
11	25125083	Nguyễn Đình Minh Huy	Coded concrete playable characters (Mario, Luigi, Toad, Peach) with custom physics/abilities	2.5	see docs/ Group52_04/52.md
12	25125083	Nguyễn Đình Minh Huy	Built the Item base class and 12 concrete item subclasses (Mushroom, FireFlower, Coin, Star, etc.)	3	see docs/ Group52_04/52.md
13	25125084	Trần Gia Huy	Built InputManager (Command Pattern) with dual-player key mapping (P1 WASD/Space/F, P2 Arrows/M)	2	see docs/ Group52_04/52.md

No	Student ID	Full name	Task Description	Hours	Evidence
14	25125084	Trần Gia Huy	Implemented SoundManager (Singleton) with music/SFX channels, WAV/OGG loading	1.5	see docs/ Group52_04/52.md
15	25125084	Trần Gia Huy	Researched/designed the IMovementStrategy interface and 8 enemy AI strategies	2	see docs/ Group52_04/52.md
16	25125084	Trần Gia Huy	Prepared abstract skeletons for the Enemy and Block type hierarchies	1	see docs/ Group52_04/52.md
17	25125084	Trần Gia Huy	Designed the EntityFactory registry blueprint for JSON-driven instantiation	1	see docs/ Group52_04/52.md
18	25125083	Nguyễn Đình Minh Huy	Implemented the full Camera class (LERP tracking, screen shake, viewport clamping)	2	see docs/ Group52_05/52.md
19	25125083	Nguyễn Đình Minh Huy	Refactored move/run commands to intent flags on Character/Player instead of direct velocity changes	1.5	see docs/ Group52_05/52.md
20	25125083	Nguyễn Đình Minh Huy	Centralized friction/acceleration/run-speed logic into PhysicsEngine::update() with surface-specific properties	2	see docs/ Group52_05/52.md
21	25125083	Nguyễn Đình Minh Huy	Integrated Camera into PlayingState with dual-view (world vs HUD) rendering	1.5	see docs/ Group52_05/52.md
22	25125083	Nguyễn Đình Minh Huy	Added mid-air wall-slide friction with a capped slide speed (WALL_SLIDE_SPEED)	1.5	see docs/ Group52_05/52.md
23	25125083	Nguyễn Đình		2.5	see docs/ Group52_05/52.md

No	Student ID	Full name	Task Description	Hours	Evidence
		Minh Huy	Integrated nlohmann/json and built LevelLoader to parse tile/entity/spawn data from level JSON		
24	25125083	Nguyễn Đình Minh Huy	Authored 4 level files (level_1..3, bonus_1) and an ImGui level selector for playtesting	2	see docs/ Group52_05/52.md
25	25125084	Trần Gia Huy	Implemented IMovementStrategy and 8 concrete AI strategies (Patrol, Chase, Fly, etc.)	2.5	see docs/ Group52_05/52.md
26	25125084	Trần Gia Huy	Coded the Enemy base class plus Goomba, KoopaTroopa, KoopaParatroopa, Boo	3	see docs/ Group52_05/52.md
27	25125084	Trần Gia Huy	Coded the Block base class plus BrickBlock, QuestionBlock, Pipe, Flagpole	2.5	see docs/ Group52_05/52.md
28	25125084	Trần Gia Huy	Coded 7 advanced v2.0 enemies (PiranhaPlant, BulletBill, HammerBro, Thwomp, ChainChomp, Lakitu, Spiny)	3.5	see docs/ Group52_05/52.md
29	25125084	Trần Gia Huy	Coded 5 advanced v2.0 blocks (HiddenBlock, MovingPlatform, FallingPlatform, IceBlock, ConveyorBelt)	2.5	see docs/ Group52_05/52.md
30	25125084	Trần Gia Huy	Implemented EntityFactory to instantiate 25+ entity types from JSON	2.5	see docs/ Group52_05/52.md
31	25125084	Trần Gia Huy	Built SpriteSheet, Animation (flyweight) and AnimationManager/Animator	3	see docs/ Group52_05/52.md
32	25125084	Trần Gia Huy	Wrote verify_enemies_new.cpp / verify_blocks_new.cpp test harnesses	1.5	see docs/ Group52_05/52.md
33	25125083	Nguyễn Đình		2.5	see docs/ Group52_06/52.md

No	Student ID	Full name	Task Description	Hours	Evidence
		Minh Huy	Built Serializer for save-slot JSON schemas (character, player state, scores, checkpoints, playtime)		
34	25125083	Nguyễn Đình Minh Huy	Implemented StatisticsTracker (EventBus subscriber for coins/enemies/deaths/combos/playtime)	1.5	see docs/ Group52_06/52.md
35	25125083	Nguyễn Đình Minh Huy	Implemented AchievementManager (10 achievements) plus a slide-in/out toast HUD	2	see docs/ Group52_06/52.md
36	25125083	Nguyễn Đình Minh Huy	Built the ImGui Pause/Settings panel (save slots, volume, keybindings, difficulty, auto-save)	2	see docs/ Group52_06/52.md
37	25125083	Nguyễn Đình Minh Huy	Wrote the verify_save_load.cpp integration test suite	1	see docs/ Group52_06/52.md
38	25125083	Nguyễn Đình Minh Huy	Implemented the Undo/Redo command framework (EditorCommands)	2	see docs/ Group52_06/52.md
39	25125083	Nguyễn Đình Minh Huy	Built the MapEditor viewport dashboard (tile/entity brushes, undo/redo, save)	2	see docs/ Group52_06/52.md
40	25125083	Nguyễn Đình Minh Huy	Implemented pixel-to-grid coordinate mapping and grid-line rendering	1.5	see docs/ Group52_06/52.md
41	25125083	Nguyễn Đình Minh Huy	Centralized duplicate tile/entity name mappers into SerializationUtils	1	see docs/ Group52_06/52.md

No	Student ID	Full name	Task Description	Hours	Evidence
42	25125083	Nguyễn Đình Minh Huy	Wrote the verify_map_editor.cpp test harness	1	see docs/ Group52_06/52.md
43	25125084	Trần Gia Huy	Participated in architectural reviews, resolved cross-branch merge conflicts, refined achievement SFX triggers	1	see docs/ Group52_06/52.md
44	25125083	Nguyễn Đình Minh Huy	Merged A/feature/save-load and A/feature/level-editor; addressed PR review fixes (EventBus sentinel, debug key remap, Game keybinding encapsulation)	2	see docs/ Group52_07/52.md
45	25125083	Nguyễn Đình Minh Huy	Fixed a stale achievement-toast bug on slot reload; aligned achievement keys with SPEC.md	1	see docs/ Group52_07/52.md
46	25125083	Nguyễn Đình Minh Huy	Built the consolidated verify_all.cpp test runner combining 6 test suites; updated CMakeLists.txt	2	see docs/ Group52_07/52.md
47	25125083	Nguyễn Đình Minh Huy	Authored two_player_ai_plan.md (2-player + Shadow Mario architecture design doc)	2.5	see docs/ Group52_07/52.md
48	25125083	Nguyễn Đình Minh Huy	Ran an interactive binary demo to validate save/load and level-editor integration	1	see docs/ Group52_07/52.md
49	25125084	Trần Gia Huy	Entity Factory & concrete entities merged into dev (integration of prior work)	0.5	see docs/ Group52_07/52.md
50	25125084	Trần Gia Huy	Enemy hierarchy + AI strategies merged into dev (integration of prior work)	0.5	see docs/ Group52_07/52.md

No	Student ID	Full name	Task Description	Hours	Evidence
51	25125084	Trần Gia Huy	Blocks/platforms merged into dev (integration of prior work)	0.5	see docs/ Group52_07/52.md
52	25125084	Trần Gia Huy	Built Hud (lives/score/coins/time/ stage) plus a boss health bar and retro typography (PressStart2P/ SuperMario256 fonts)	2	see docs/ Group52_07/52.md
53	25125084	Trần Gia Huy	Finalized SpriteSheet/ AnimationManager/Animator; added verify_blocks/verify_enemies/ verify_graphics/verify_graphics_visual/ verify_hud_boss test targets	1.5	see docs/ Group52_07/52.md
54	25125084	Trần Gia Huy	Consolidated duplicate asset/ and assets/ folders into one assets/ root	0.5	see docs/ Group52_07/52.md
55	25125084	Trần Gia Huy	Built the Minimap component (scaled tilemap preview, camera frustum, player tracking dot)	2	Git commit cb6eab2
56	25125084	Trần Gia Huy	Wrote the verify_minimap_visual.cpp test target	1	see docs/ Group52_07/52.md
57	25125083	Nguyễn Đình Minh Huy	Reviewed/validated the ParticleSystem/ObjectPool design (O(1) acquire/release, zero-allocation)	1	see docs/ Group52_08/52.md
58	25125083	Nguyễn Đình Minh Huy	Updated TASKS.md through Phase 8 and Bonus A, cross-referenced against TASK_DIVISION.md	0.5	see docs/ Group52_08/52.md
59	25125083	Nguyễn Đình Minh Huy	Ran git fetch --all, verified dev in sync with origin/dev, tracked branch status	0.5	see docs/ Group52_08/52.md
60	25125084			2.5	

No	Student ID	Full name	Task Description	Hours	Evidence
		Trần Gia Huy	Implemented ParticleSystem with ObjectPool (burst/continuous emission modes)		Git commit 7a6f62b
61	25125084	Trần Gia Huy	Tuned particle parameters (alpha fade, emission rate, spread angle)	1.5	Git commits c7d4fd4, 62c78ef
62	25125084	Trần Gia Huy	Built the verify_particles_visual interactive test target	1	see docs/Group52_08/52.md
63	25125084	Trần Gia Huy	Extracted the player sprite sheet player.json (92 animation frames: idle/walk/run/jump/etc.)	3	Git commits d6bd07d, 0be4508, 9403be5
64	25125084	Trần Gia Huy	Built the verify_player_animation_visual test target	1	see docs/Group52_08/52.md
65	25125084	Trần Gia Huy	Began sprite naming standardization / asset pipeline scripts	1	Git commit 15327b4
66	25125084	Trần Gia Huy	Enhanced random-number generation with std::uniform_real_distribution for particle spread	0.5	Git commit 9bb9d16
67	25125083	Nguyễn Đình Minh Huy	Implemented the Fireball entity + ObjectPool, EntityFactory/EventBus wiring, verify_fireball.cpp (4/4 pass)	3	see docs/Group52_09/52.md
68	25125083	Nguyễn Đình Minh Huy	Enhanced Flagpole with height-scaled scoring, animation and a LevelComplete event; verify_flagpole_trampoline.cpp	2	see docs/Group52_09/52.md
69	25125083	Nguyễn Đình Minh Huy	Enhanced Trampoline bounce mechanics (spring compression, configurable impulse)	1.5	see docs/Group52_09/52.md
70	25125083			2	

No	Student ID	Full name	Task Description	Hours	Evidence
		Nguyễn Đình Minh Huy	Built the CheckpointFlag entity (visual states, auto-save, CheckpointActivated event, respawn coordinates)		see docs/ Group52_09/52.md
71	25125083	Nguyễn Đình Minh Huy	Implemented the Memento pattern: GameSnapshot + TimeRewindManager (300-frame circular buffer); verify_memento_rewind.cpp (4/4 pass)	3.5	see docs/ Group52_09/52.md
72	25125083	Nguyễn Đình Minh Huy	Built MapGenerator for procedural level generation; verify_map_generator.cpp	3	see docs/ Group52_09/52.md
73	25125083	Nguyễn Đình Minh Huy	Implemented the dual bounding-box architecture (physicsBox/combatBox) across Entity/CollisionDetector/CollisionResolver	2.5	see docs/ Group52_09/52.md
74	25125083	Nguyễn Đình Minh Huy	Added void-fall death detection in PlayingState	1	see docs/ Group52_09/52.md
75	25125083	Nguyễn Đình Minh Huy	Added brick/question-block head-butt reactions in PhysicsEngine	1.5	see docs/ Group52_09/52.md
76	25125083	Nguyễn Đình Minh Huy	Removed public setLives/setCoins/setScore from Player; added friend class PlayingState	0.5	see docs/ Group52_09/52.md
77	25125083	Nguyễn Đình Minh Huy	Added Camera::move() for editor WASD/arrow panning	1	see docs/ Group52_09/52.md
78	25125083	Nguyễn Đình	Added a Level Editor Mode (F1) button to MenuState	0.5	see docs/ Group52_09/52.md

No	Student ID	Full name	Task Description	Hours	Evidence
		Minh Huy			
79	25125083	Nguyễn Đình Minh Huy	Hardened Game::shutdown() to pop all game states before manager teardown	1	see docs/ Group52_09/52.md
80	25125083	Nguyễn Đình Minh Huy	Added multi-path font-loading fallback candidates in PlayingState	0.5	see docs/ Group52_09/52.md
81	25125083	Nguyễn Đình Minh Huy	Fetches/merged origin/B/feature/ graphic-visual into dev; resolved font-loading + agent_history.log conflicts; pushed dev	1	Git commit 17cef23
82	25125083	Nguyễn Đình Minh Huy	Fixed non-solid TileType::Coin tiles not being collected on walk-through	1	Git commit d12e763
83	25125083	Nguyễn Đình Minh Huy	Deduplicated a ResourceManager missing-SoundBuffer warning spam	0.5	see docs/ Group52_09/52.md
84	25125084	Trần Gia Huy	Built a spritesheet-merging tool; updated enemy_projectile/item/world_scenery_item atlases	3	Git commits 044a5d8, 80c048e, b90a899
85	25125084	Trần Gia Huy	Cleaned sprite/atlas coordinates affected by the merge	1	Git commit 10828a7
86	25125084	Trần Gia Huy	Updated SPRITE_NAMING_CONVENTIONS.md (frame inventories, naming standards)	1	see docs/ Group52_09/52.md
87	25125084	Trần Gia Huy	Built SpriteColorFilter (HSL->RGB, 12Hz Star Power cycling, 15Hz hurt flicker, hit-flash tint)	2	Git commit 2c3bb83

No	Student ID	Full name	Task Description	Hours	Evidence
88	25125084	Trần Gia Huy	Built SpriteTransformAnim (scale-lerp easing, 720 deg/s spin, 12Hz Mega/Mini flicker)	2	Git commit 2c3bb83
89	25125084	Trần Gia Huy	Built the EntityDeathEffect singleton (EnemyFlip, StarKillSpin, PlayerDeathHop)	2	Git commit 2c3bb83
90	25125084	Trần Gia Huy	Added a SoundManager synthesized fallback beep buffer	1	see docs/Group52_09/52.md
91	25125084	Trần Gia Huy	Upgraded verify_graphics_visual with a Transform/Color-Filter FX ImGui tab	1.5	see docs/Group52_09/52.md
92	25125083	Nguyễn Đình Minh Huy	Refined MapGenerator (elevation profiles, biome ceilings, lava pits, jump-reachability-guarded platforms, 3 Star Coins, threat-pacing curve) + ImGui tuning panel	3.5	see docs/Group52_10/52.md
93	25125083	Nguyễn Đình Minh Huy	Built the 3-world campaign + warp-pipe system; generate_game_levels.cpp synthesizing 7 level JSONs; extended SerializationUtils/LevelLoader for warp metadata; seamless sub-level transitions	4	see docs/Group52_10/52.md
94	25125083	Nguyễn Đình Minh Huy	Added ImGui campaign navigation dropdowns and an Active Level Tube warp selector	1.5	see docs/Group52_10/52.md
95	25125083	Nguyễn Đình Minh Huy	Fixed a static-block gravity bug (getGravityMultiplier() override); added item resting-state physics (resolveItemVsBlock)	1.5	see docs/Group52_10/52.md
96	25125083	Nguyễn Đình	Optimized player hitbox dimensions (32px to 24px width tiers); refined stomp-vs-side-hit combat resolution	2	see docs/Group52_10/52.md

No	Student ID	Full name	Task Description	Hours	Evidence
		Minh Huy			
97	25125083	Nguyễn Đình Minh Huy	Fixed the LevelLoader ifstream failbit cascade (file.clear()); expanded fallback search paths	1.5	see docs/ Group52_10/52.md
98	25125083	Nguyễn Đình Minh Huy	Fixed a SIGABRT shutdown crash (explicit m_window.close()) and hardened GameStateManager transitions	1.5	see docs/ Group52_10/52.md
99	25125084	Trần Gia Huy	Wired sprite animations across all entities; aspect-ratio scaling for HUD/items; POW 8-frame shockwave; trampoline/PSwitch visual states	3.5	Git commits 069817b, 0bc21c3, 2d5058e, 83660b5, 065557d, 61e30bc
100	25125084	Trần Gia Huy	Built the Web Metadata Editor Studio (metadata_editor.html, edit_sprite_metadata.py) with cursor-anchored zoom, position-finder HUD, drag-to-draw frames, TexturePacker-compatible export	2.5	see docs/ Group52_10/52.md
101	25125084	Trần Gia Huy	Integrated sprite HUD assets (animated coin counter, Star Coin indicators, character-specific lives icon)	1.5	see docs/ Group52_10/52.md
102	25125084	Trần Gia Huy	Built PipeRenderer (2-block-wide AABB, directional multi-part pipe head/body rendering)	1.5	Git commit a0f1a3b
103	25125084	Trần Gia Huy	Wired game-wide SFX/BGM into SoundManager (jump, stomp, coin, break, power-up/down, boing, stage-clear, game-over) + fallback buffer	2.5	Git commits 1002595, c383c19, 9976209
104	25125084	Trần Gia Huy	Tuned all 11 enemy AI state machines (Thwomp slam cycle, Boo gaze-detection, HammerBro parabolic throw, Koopa shell physics)	3	Git commit d954a66

No	Student ID	Full name	Task Description	Hours	Evidence
105	25125084	Trần Gia Huy	Built verify_enemies_behavior.cpp + enhanced verify_enemies_visual.cpp (11 isolated test rooms)	1.5	Git commit d954a66
106	25125083	Nguyễn Đình Minh Huy	Resolved the Level 1-3 crash on Bowser's first fireball (mid-frame entity-spawn vector reallocation); rebuilt and playtested on Windows	3	see logs/ agent_history.log 2026-08-16/22
107	25125083	Nguyễn Đình Minh Huy	Fixed 12 defects from a playtest pass plus two menus that overflowed	2	Git commit f6fc8bc
108	25125083	Nguyễn Đình Minh Huy	Fixed CMakeLists naming src/Graphics/HUD.cpp when the file is Hud.cpp	0.5	Git commit 03bf828
109	25125083	Nguyễn Đình Minh Huy	Fixed the player being unable to jump while walking into a wall	1	Git commit 60cc735
110	25125083	Nguyễn Đình Minh Huy	Fixed 28 files using std:: symbols without including their header	1	Git commit 4ac8a1c
111	25125083	Nguyễn Đình Minh Huy	Fixed CI building only 2 of the 13 binaries it then tested	1	Git commit 8141120
112	25125083	Nguyễn Đình Minh Huy	Fixed 4 test harnesses opening an sf::RenderWindow on a CI runner with no display	1	Git commit 33d5aae
113	25125083	Nguyễn Đình Minh Huy	Fixed three defects the screenshots showed that the tests did not (backdrop ground line, hill-bottom	2	Git commit da8ca99

No	Student ID	Full name	Task Description	Hours	Evidence
			offset, a Player 2 HUD badge overwriting the world field)		
114	25125083	Nguyễn Đình Minh Huy	Sealed test hermeticity: Serializer::setSaveDirectory() + TestSaveSandbox.hpp so all 23 verify_* harnesses point at a fresh temp directory	2.5	Git commit 045f9b7
115	25125083	Nguyễn Đình Minh Huy	Generated real UML class diagrams from the headers (tools/gen_class_diagram.py) and wrote the CS202 final report's first edition (15 sections, generated by build_report.py)	4	Git commits a0395d6, 3c0a768, 64088b4
116	25125083	Nguyễn Đình Minh Huy	Produced a LaTeX/PDF edition of the report in the HCMUS template and deepened it (implementation section 3->11 subsections, technical-problems section rebuilt into 8 themed groups)	4	Git commits fdec9cc, 730336c
117	25125083	Nguyễn Đình Minh Huy	Diagnosed and fixed three defects reported from a real playthrough: the end-of-level castle's buffer margin, Boom Boom's arena overlapping the 1-2 flagpole, and the camera/z-order bug carried into 1-3; added regression tests for all three	4	this session, branch A/fix/level-completion-and-camera-defects
118	25125083	Nguyễn Đình Minh Huy	Implemented Endless Mode (chunked infinite level generation with rising difficulty, distance-based scoring) and a real solvability oracle for MapGenerator, after evaluating (and rejecting the framework of) the abandoned A/mapgen-gan-plan and A/rl-neural-policy side branches for anything worth porting	5	this session, branch A/feature/endless-mode-and-solvability
119	25125083	Nguyễn Đình Minh Huy	Rebuilt the submission package: a 97-feature list and per-member manual test checklist grounded in a fresh	4	this session, branch A/docs/

No	Student ID	Full name	Task Description	Hours	Evidence
		Minh Huy	source-code audit, the Member Contributions spreadsheet (matching the DesignPatternsGroup52 format), and an expanded report (OOP/pattern rationale, two new UML diagrams)		submission-package-update
120	25125084	Trần Gia Huy	Resolved merge conflicts while pulling the latest changes and merged into main	2	Git commits 72b13c6, a522007, 75e9cbb